

RAGHAVENDRA DINESH

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SUMMARY

Data Scientist with a robust background in machine learning and advanced analytics, skilled in Python, SQL, and data visualization tools. Successfully engineered AI-driven applications that optimize decision-making and mitigate risk. Brings strong analytical and problem-solving abilities to develop fraud detection strategies and data-driven solutions.

TECHNICAL SKILLS

- **Programming & Scripting:** Python, C & Embedded C, MATLAB/Simulink, C++, Java, JavaScript, HTML, Bash
- **Interests:** Computer Vision, Deep Learning, Natural Language Processing, Generative AI
- **Tools:** Pandas, NumPy, Tensorflow, Keras, PyTorch, OpenCV2, Sklearn, TensorRT, Onnx, TFLite, SQL, GIT, Docker
- **Data Science & Analytics:** Machine Learning, Data Visualization, Statistical Analysis, Analytical Skills

PROFESSIONAL EXPERIENCE

Indra Solutions LLC | Data Scientist

May 2025 - Present

- Engineered an app using Python-based FastAPI and LLMs to analyze contract data and profile details, resulting in more efficient generation of RFP documents
- Expedited proposal submissions, enabling the team to target and secure contracts ahead of competitors

Arizona State University | Deep Learning Engineer

Jun 2024 - May 2025

- Engineered a custom BLIP-2 that integrates large vision and language models for plant health diagnostics by leveraging machine learning techniques, attaining 98% accuracy with a Lite ResNet50 model and 96% accuracy with a fine-tuned SWIN Transformer over 40 epochs
- Enhanced diagnostic precision by developing an end-to-end solution that leverages QFormer to interface a frozen SWIN encoder with Llama 3.2 3B, achieving a maximum RougeL F1 score of 0.86 for generating localized cure
- Improved treatment insights by first validating a proof-of-concept using Gemini API (RougeL F1 score of 0.2) and later optimizing performance with PEFT on Vicuna 7B (score of 0.6) before transitioning to Llama 3.2 3B

America Reads, Mary Lou Fulton Teachers College | Tutor

Oct 2022 - May 2024

- Developed a Gemini-driven platform that automatically generated daily activity plans and personalized worksheets, improving student engagement and learning outcomes
- Enhanced adaptive assessment by designing a module that customizes questions based on student progress, leading to more accurate evaluations and tailored learning experiences

Bosch Global Software Technologies Pvt. Ltd. | Automation Engineer

Sep 2019 - Aug 2022

- Delivered an RF emission analysis portal with interactive visualizations using Plotly, applying agile practices during iterative refinements of an Autoencoder anomaly detection system to reduce analysis time from 30–50 hours to 6–8 hours
- Advanced lab asset management by designing a smart inventory tracking system on Raspberry Pi 3B that integrated YOLOv4-tiny for asset detection and Tesseract OCR for auto labeling, providing real-time tracking
- Accelerated report generation by building a web application to automate data extraction and document rendering, reducing manual processing time by 90%

ConnectM Technology Solutions Pvt. Ltd. | Artificial Intelligence Intern

Jan 2019 - Mar 2019

- Developed a Driver Assistance System with safety constraints using facial recognition and voice assistance, enhancing driver safety by alerting them to potential hazards
- Built voice and face recognition systems using CMU Pocket Sphinx, Haar Cascade, and AWS Face Rekognition, improving system accuracy and user interaction
- Trained a drowsiness detection model using facial landmark detection and integrated all modules on Raspberry Pi 3B, resulting in a comprehensive system that alerts drivers to prevent accidents

EDUCATION

Arizona State University

Masters, Artificial Intelligence (Robotics and Autonomous Systems) (GPA: 3.97 GPA)

PES Institute of Technology

Bachelor of Engineering, Electronics and Communication Engineering (GPA: 3.20 GPA)

PROJECTS

Negative CLIPort: Robotic Arm enhanced with Negative Goals | Perception in Robots

2023

- Improved CLIPort to handle a wider range of requirements by making it more robust to negative language goals.
- Training data made with actions the arm avoids, and loss function modified to penalize conflicting actions.

Medical Image Analysis for Lung Disease Detection | Medical Image Processing

2023

- Leveraged Swin Transformer, Faster R-CNN, UPerNet, and U-Net for classification, localization, and segmentation of Chest X Rays from VinBigData CXR, SIIM-ACR Pneumothorax, MedMNISTv2 and MINISRT
- Developed a multi-task model to accurately analyze X-rays, achieving high score in metrics like AUC, IoU, FROC.

CERTIFICATION

- **MLOps | Machine Learning Operations:** Duke University, Jul 2023 | MLflow, DevOps, AWS SageMaker, MLOps
- **Generative AI with Large Language Models:** DeepLearning.AI, Sep 2024 | SageMaker, LLMs, PEFT, Quantization
- **DeepLearning.AI TensorFlow Developer:** DeepLearning.AI, Jul 2024 | Keras, Algorithms, OpenCV
- **Deep Learning Specialization:** DeepLearning.AI, May 2020 | Image Processing, Data Analysis, Neural Networks
- **Machine Learning:** DeepLearning.AI, Nov 2019 | Applied Mathematics, Machine Learning, Pattern Recognition